

## SAT Math Mastery · Week TP1C

## Trick Pack 1 — Part C

**Time** 120 minutes: 90 min solving + 30 min self-check and error log

**Topics** exponent rewriting · mixture balance · parallel sides in polygons · similar triangles · discriminant conditions · composite functions · coefficient parity · quadratic modeling · trig identities · exponential ratios

**Name / Date** \_\_\_\_\_

## 1 · SKILLS &amp; IDEAS

- S1.** Factor composite bases into primes, combine like powers, and shift an exponent so a chosen function value becomes the coefficient.
- S2.** Track one component through a multi-step mixing process with a single balance equation.
- S3.** Use parallel sides (supplementary co-interior angles) with the polygon angle sum, and angle bisectors in a trapezoid.
- S4.** Use similar triangles together with  $(a + b)^2 = a^2 + b^2 + 2ab$  to avoid solving a quadratic.
- S5.** Turn "exactly one solution", "tangent", or "one  $x$ -intercept" into discriminant  $= 0$ , then apply the stated restriction.
- S6.** Build quadratics from vertex, roots, or data; evaluate composites by substituting the target input first.
- S7.** Reciprocal pairs  $b^x, b^{-x}$  and positivity of exponentials; constant-ratio exponentials and percent change.
- S8.** Prove or refute "must be" claims about coefficients with exhaustive parity or divisibility cases.
- S9.** Complementary angles:  $\sin(90^\circ - \theta) = \cos \theta$ , and sums versus products of ratios.

## 2 · REQUIRED FACTS

- $b^x > 0$  for every real  $x$  when  $b > 0$ ;  $b^{-x} = \frac{1}{b^x}$ ;  $b^{x+k} = b^k \cdot b^x$ .
- $ax^2 + bx + c = 0$  has exactly one real solution  $\iff b^2 - 4ac = 0$ ; a line is tangent to a parabola  $\iff$  the equated quadratic has discriminant 0.
- Co-interior (same-side) angles between parallel lines sum to  $180^\circ$ ; interior angles of an  $n$ -gon sum to  $(n-2) \cdot 180^\circ$ .
- Angles in a triangle sum to  $180^\circ$ ; a line parallel to a side of a triangle cuts off a similar triangle.
- If  $\alpha + \beta = 90^\circ$  then  $\sin \alpha = \cos \beta$  and  $\tan \alpha = \frac{1}{\tan \beta}$ ;  $\sin^2 \theta + \cos^2 \theta = 1$ .
- A percent decrease of  $p\%$  per step means a constant ratio  $b = 1 - \frac{p}{100}$ .

## 3 · REQUIRED FORMULAS

$$f(x) = a(x - h)^2 + k, \quad f(x) = a(x - r_1)(x - r_2), \quad \Delta = b^2 - 4ac, \quad (u + v)^2 = u^2 + v^2 + 2uv$$

$$\text{amount} = \text{fraction} \times \text{mass}, \quad \tan \theta + \cot \theta = \frac{1}{\sin \theta \cos \theta}, \quad f(x) = f(x_0) b^{x-x_0}$$

## 4 · MINI THEORY SHEET

**Rewrite, then read.** Factor every base into primes first; collect each prime's exponent; a single coefficient times a single base is the target form.

**Parameter conditions.** *One solution / tangent / one  $x$ -intercept*  $\Rightarrow \Delta = 0$ . Solve the resulting equation in the parameter and then use the sign or size restriction stated in the question.

**Composite functions.** Never expand. Compute the inner value at the target input, then feed it to the outer function.

**Speed shortcuts.**

| Situation                                 | Shortcut                                                          |
|-------------------------------------------|-------------------------------------------------------------------|
| composite bases in a power                | prime-factor every base; one coefficient times one base           |
| want $f(x_0)$ as the coefficient          | $f(x) = f(x_0) b^{x-x_0}$                                         |
| mixing with added pure component          | one balance: original + added = fraction $\times$ final total     |
| parallel sides in a polygon               | co-interior angles sum to $180^\circ$ ; then use $(n-2)180^\circ$ |
| inscribed square in a right triangle      | $s = \frac{ab}{a+b}$ and $(a+b)^2 = c^2 + 4 \cdot \text{Area}$    |
| one real root / tangent                   | $\Delta = 0$ ; the sign condition picks the root                  |
| coefficient from two points               | add and subtract equations to isolate $b$ and $a+c$               |
| vertex plus one point                     | $a(x-h)^2 + k$ ; composite: inner first                           |
| $b^p$ and $b^{-p}$ both given             | set $t = b^p$ , subtract, keep $t > 0$                            |
| "must be true" about integer coefficients | list all factor pairs; check parity or remainder in each          |
| $\sin \theta + \cos \theta$ given         | square it for $\sin \theta \cos \theta$                           |
| constant percent change                   | divide values to cancel the coefficient; shift is irrelevant      |

**5 · GUIDED EXAMPLES****Example 1 — Shift the exponent to expose a value**

$f(x) = \frac{4^x \cdot 6^{x+1}}{9^x}$ . Write  $f$  so that  $f(3)$  is the coefficient.

$$4^x = 2^{2x}, 6^{x+1} = 2^{x+1} 3^{x+1}, 9^x = 3^{2x}, \text{ so } f(x) = 2^{3x+1} 3^{1-x} = 6 \cdot \left(\frac{8}{3}\right)^x.$$

$$f(3) = 6 \cdot \frac{512}{27} = \frac{1024}{9}, \text{ hence } f(x) = \frac{1024}{9} \left(\frac{8}{3}\right)^{x-3}.$$

**Example 2 — Tangent line with a parameter**

For which positive  $k$  is  $y = kx - 1$  tangent to  $y = x^2 + 3x + 3$ ?

$$x^2 + 3x + 3 = kx - 1 \Rightarrow x^2 + (3-k)x + 4 = 0. \Delta = (3-k)^2 - 16 = 0 \Rightarrow k = 7 \text{ or } k = -1. \text{ Positive: } k = 7.$$

**Example 3 — A sum gives a product, then a reciprocal**

For acute  $\theta$ ,  $\sin \theta + \cos \theta = \frac{\sqrt{7}}{2}$ . Find  $\tan \theta + \tan(90^\circ - \theta)$ .

$$\text{Square: } 1 + 2 \sin \theta \cos \theta = \frac{7}{4} \Rightarrow \sin \theta \cos \theta = \frac{3}{8}. \text{ Then } \tan \theta + \cot \theta = \frac{1}{\sin \theta \cos \theta} = \frac{8}{3}.$$

## 6 · PRACTICE SET — 15 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

SAT CHALLENGE · CALCULATOR

The function  $f$  is defined by  $f(x) = \frac{9^x \cdot 12^{x+1}}{8^x}$ . Which of the following expressions is equivalent to  $f(x)$  and displays the value of  $f(2)$  as a coefficient?

(A)  $\frac{729}{4} \left(\frac{27}{2}\right)^{x-2}$

(B)  $2187 \left(\frac{27}{2}\right)^{x-2} \checkmark$

(C)  $2187 \left(\frac{27}{2}\right)^{x+2}$

(D)  $162 \left(\frac{27}{2}\right)^{x-1}$

1. Prime factors:  $9^x = 3^{2x}$ ,  $12^{x+1} = 2^{2x+2} \cdot 3^{x+1}$ ,  $8^x = 2^{3x}$ .2.  $f(x) = 3^{2x+x+1} \cdot 2^{2x+2-3x} = 3^{3x+1} \cdot 2^{2-x} = 12 \left(\frac{27}{2}\right)^x$ .3.  $f(2) = 12 \cdot \frac{729}{4} = 2187$ .4. Shift the exponent:  $f(x) = f(2) \left(\frac{27}{2}\right)^{x-2} = 2187 \left(\frac{27}{2}\right)^{x-2}$ .5. Other options:  $162 \left(\frac{27}{2}\right)^{x-1}$  is equivalent but shows  $f(1)$ ;  $\frac{729}{4} \left(\frac{27}{2}\right)^{x-2}$  and the  $x+2$  form are not equivalent.**Answer: (B)**

2.

SAT CHALLENGE · CALCULATOR

A metalworker melts together 150 g of alloy X, which is 20% nickel by mass, and  $m$  g of alloy Y, which is 50% nickel by mass. Then 40 g of pure nickel is stirred into the melt. The final alloy is 40% nickel by mass. What is the value of  $m$ ?

(A) 250

(B) 460

(C) 60  $\checkmark$

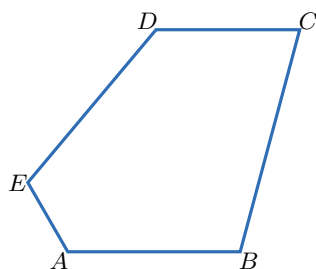
(D) 100

1. Final mass:  $150 + m + 40 = 190 + m$ , and nickel is 40% of it:  $0.4(190 + m)$ .2. Nickel in:  $0.20(150) + 0.50m + 40 = 70 + 0.5m$ .3. Balance:  $70 + 0.5m = 76 + 0.4m \Rightarrow 0.1m = 6 \Rightarrow m = 60$ .4. Check: total 250 g, nickel  $30 + 30 + 40 = 100$  g, and  $100/250 = 40\%$ .**Answer: (C)**

3.

SAT CHALLENGE · NO CALCULATOR NEEDED

In convex pentagon  $ABCDE$  shown (not drawn to scale), side  $\overline{AB}$  is parallel to side  $\overline{DC}$ . The measures of four interior angles are  $\angle A = (6x)^\circ$ ,  $\angle B = (5x + 5)^\circ$ ,  $\angle C = (3x + 15)^\circ$  and  $\angle D = (7x - 10)^\circ$ . What is the measure, in degrees, of  $\angle E$ ?



(A) 110 ✓

(B) 70

(C) 105

(D) 130

1.  $\overline{AB} \parallel \overline{DC}$  with transversal  $\overline{BC}$ :  $\angle B + \angle C = 180^\circ$  (co-interior).

2.  $(5x + 5) + (3x + 15) = 180 \Rightarrow 8x = 160 \Rightarrow x = 20$ .

3.  $\angle A = 120^\circ$ ,  $\angle B = 105^\circ$ ,  $\angle C = 75^\circ$ ,  $\angle D = 130^\circ$ .

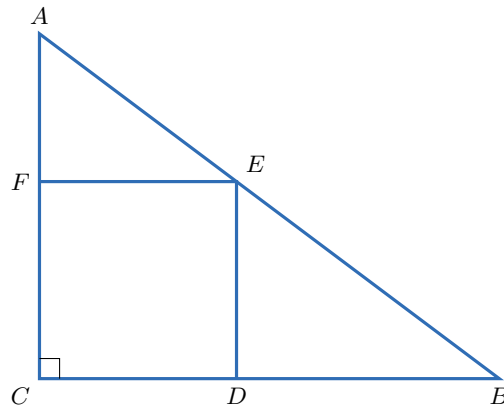
4. Pentagon angle sum  $= (5 - 2) \cdot 180^\circ = 540^\circ$ .

5.  $\angle E = 540 - (120 + 105 + 75 + 130) = 110^\circ$ .

**Answer: (A)**

4. SAT CHALLENGE · NO CALCULATOR NEEDED

In right triangle  $ABC$ ,  $\angle C = 90^\circ$ ,  $AB = 20$ , and the area of the triangle is 96. Square  $CDEF$  is inscribed in the triangle, with  $D$  on  $\overline{BC}$ ,  $F$  on  $\overline{AC}$ , and  $E$  on  $\overline{AB}$ . What is the side length of the square?



(A)  $\frac{24}{7}$

(B)  $\frac{48}{5}$

(C)  $\frac{96}{7}$

(D)  $\frac{48}{7}$  ✓

1. Let the legs be  $a = BC$ ,  $b = AC$ . Area:  $\frac{1}{2}ab = 96 \Rightarrow ab = 192$ . Pythagorean:  $a^2 + b^2 = 400$ .

2.  $(a + b)^2 = a^2 + b^2 + 2ab = 400 + 384 = 784 \Rightarrow a + b = 28$  (no quadratic needed).

3. Triangle  $FAE \sim$  triangle  $CAB$  (since  $\overline{FE} \parallel \overline{CB}$ ):  $\frac{AF}{AC} = \frac{FE}{CB}$ , i.e.  $\frac{b-s}{b} = \frac{s}{a}$ .

4. This gives  $s \left( \frac{1}{a} + \frac{1}{b} \right) = 1$ , so  $s = \frac{ab}{a+b} = \frac{192}{28} = \frac{48}{7}$ .

5. (Legs are 12 and 16;  $\frac{12 \cdot 16}{28} = \frac{48}{7}$ .)

**Answer: (D)**

5.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function  $f$  is defined by  $f(x) = ax^2 + bx + c$ , where  $a$ ,  $b$ , and  $c$  are constants. The graph of  $y = f(x)$  in the  $xy$ -plane passes through  $(1, 8)$  and  $(-1, 2)$  and has exactly one  $x$ -intercept. What is the greatest possible value of  $a$ ?

(A)  $\frac{5}{2}$

(B) 9

(C)  $\frac{9}{2}$  ✓

(D)  $\frac{1}{2}$

1.  $f(1) = a + b + c = 8$  and  $f(-1) = a - b + c = 2$ . Subtract:  $2b = 6 \Rightarrow b = 3$ . Add:  $a + c = 5 \Rightarrow c = 5 - a$ .

2. Exactly one  $x$ -intercept:  $b^2 - 4ac = 0 \Rightarrow 9 = 4a(5 - a)$ .

3.  $4a^2 - 20a + 9 = 0 \Rightarrow (2a - 1)(2a - 9) = 0 \Rightarrow a = \frac{1}{2}$  or  $a = \frac{9}{2}$ .

4. The greatest possible value is  $\frac{9}{2}$ .

Answer: (C)

6.

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function  $f$  has its vertex at  $(3, -2)$ , and  $f(5) = 6$ . The function  $g$  is defined by  $g(x) = \frac{3}{4}f\left(\frac{1}{2}x - 1\right)$ .

The graph of  $y = f(x)$  has  $y$ -intercept  $(0, p)$  and the graph of  $y = g(x)$  has  $y$ -intercept  $(0, q)$ . What is the value of  $q - p$ ?

(A)  $-\frac{23}{2}$

(B)  $\frac{13}{2}$  ✓

(C)  $-4$

(D)  $\frac{45}{2}$

1. Vertex form:  $f(x) = a(x - 3)^2 - 2$ . From  $f(5) = 6$ :  $4a - 2 = 6 \Rightarrow a = 2$ , so  $f(x) = 2(x - 3)^2 - 2$ .

2.  $p = f(0) = 2(9) - 2 = 16$ .

3.  $q = g(0) = \frac{3}{4}f\left(\frac{1}{2} \cdot 0 - 1\right) = \frac{3}{4}f(-1)$ , and  $f(-1) = 2(16) - 2 = 30$ , so  $q = \frac{45}{2}$ .

4.  $q - p = \frac{45}{2} - 16 = \frac{13}{2}$ .

Answer: (B)

7.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function  $f$  is defined by  $f(x) = 2^x + k$ , where  $k$  is a constant. For a real number  $p$ ,  $f(p) = 9$  and  $f(-p) = \frac{21}{4}$ . What is the value of  $p$ ?

1. Let  $t = 2^p$ , so  $t > 0$  and  $2^{-p} = \frac{1}{t}$ . Then  $t + k = 9$  and  $\frac{1}{t} + k = \frac{21}{4}$ .
2. Subtract to eliminate  $k$ :  $t - \frac{1}{t} = \frac{15}{4} \Rightarrow 4t^2 - 15t - 4 = 0 \Rightarrow (4t + 1)(t - 4) = 0$ .
3.  $t = -\frac{1}{4}$  is rejected because  $2^p > 0$ ; so  $t = 4$ .
4.  $2^p = 4 \Rightarrow p = 2$ . (Check:  $k = 5$ ;  $f(2) = 9$ ,  $f(-2) = \frac{1}{4} + 5 = \frac{21}{4}$ .)

**Answer:** 2

8.

SAT CHALLENGE · NO CALCULATOR NEEDED

For integers  $a$ ,  $b$ ,  $c$ , and  $d$ ,  $(ax + b)(cx + d) = 6x^2 + kx + 5$  for all values of  $x$ , where  $k$  is a constant. Which of the following must be true?

- I.  $k$  is odd
- II. When  $k$  is divided by 4, the remainder is 1
- III.  $k$  is not divisible by 5

(A) II and III only

(B) I, II, and III

(C) I only

**(D) I and III only ✓**

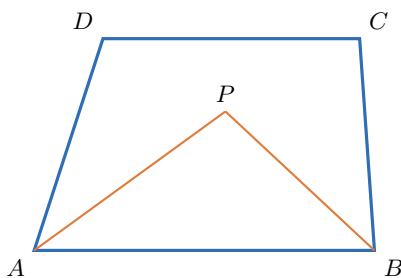
1. Matching coefficients:  $ac = 6$ ,  $bd = 5$ ,  $k = ad + bc$ .
2. Since  $bd = 5$  is odd,  $b$  and  $d$  are both odd. Since  $ac = 6$  is not divisible by 4, exactly one of  $a, c$  is even. Then  $k = ad + bc \equiv a + c \equiv 1 \pmod{2}$ . So I is true.
3. Since  $bd = 5$ , exactly one of  $b, d$  is a multiple of 5; say  $5 \mid b$ . Then  $k \equiv ad \pmod{5}$ , and  $a \mid 6$  and  $d = \pm 1$  are not multiples of 5. (Same if  $5 \mid d$ .) So III is true.
4. II:  $(a, c, b, d) = (1, 6, 5, 1)$  gives  $k = 1 \cdot 1 + 5 \cdot 6 = 31$ , remainder 3 on division by 4. So II is false, even though  $k = 13$  and  $k = 17$  satisfy it.
5. Possible values:  $k = \pm 11, \pm 13, \pm 17, \pm 31$ . Only I and III must be true.

**Answer:** (D)

9.

SAT CHALLENGE · NO CALCULATOR NEEDED

In trapezoid  $ABCD$  shown (not drawn to scale),  $\overline{AB} \parallel \overline{DC}$ ,  $\angle D = 108^\circ$ , and  $\angle C = 94^\circ$ . The bisector of  $\angle DAB$  and the bisector of  $\angle ABC$  intersect at point  $P$  inside the trapezoid. What is the measure, in degrees, of the obtuse angle formed at  $P$  by these two bisectors?



(A) 101 ✓

(B) 79

(C) 22

(D) 108

- $\overline{AB} \parallel \overline{DC}$ :  $\angle A + \angle D = 180^\circ \Rightarrow \angle A = 72^\circ$ ;  $\angle B + \angle C = 180^\circ \Rightarrow \angle B = 86^\circ$ .
- Half-angles:  $\angle PAB = 36^\circ$ ,  $\angle PBA = 43^\circ$ .
- In triangle  $APB$ :  $\angle APB = 180 - 36 - 43 = 101^\circ$ , which is obtuse.
- The vertical/supplementary angle is  $79^\circ$  (acute), so the obtuse angle is  $101^\circ$ .

**Answer: (A)**

10.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the  $xy$ -plane, the line  $y = k(x - 1) - 2$ , where  $k$  is a positive constant, is tangent to the parabola  $y = x^2 + 2x + 4$ . What is the value of  $k$ ?

(A) 8

(B) 10 ✓

(C) -2

(D) 4

- Set equal:  $x^2 + 2x + 4 = kx - k - 2 \Rightarrow x^2 + (2 - k)x + (k + 6) = 0$ .
- Tangency means exactly one solution:  $(2 - k)^2 - 4(k + 6) = 0$ .
- $k^2 - 8k - 20 = 0 \Rightarrow (k - 10)(k + 2) = 0 \Rightarrow k = 10$  or  $k = -2$ .
- Since  $k > 0$ ,  $k = 10$ . (The tangent point is  $x = \frac{k - 2}{2} = 4$ , which is not the answer.)

**Answer: (B)**

**11.**

SAT CHALLENGE · CALCULATOR

The altitude  $h$ , in meters, of a small rocket  $t$  seconds after launch is modeled by  $h(t) = at^2 + bt + c$ , where  $a$ ,  $b$ , and  $c$  are constants. At launch the altitude is 3 meters; 2 seconds after launch it is 43 meters; and 5 seconds after launch it is 28 meters. According to the model, what is the altitude, in meters, 3 seconds after launch?

1. Launch altitude:  $h(0) = c = 3$ .
2.  $h(2) = 4a + 2b + 3 = 43 \Rightarrow 2a + b = 20$  (divide by 2).
3.  $h(5) = 25a + 5b + 3 = 28 \Rightarrow 5a + b = 5$  (divide by 5).
4. Subtract:  $3a = -15 \Rightarrow a = -5$ ; then  $b = 20 - 2(-5) = 30$ .
5.  $h(3) = -5(9) + 30(3) + 3 = -45 + 90 + 3 = 48$ .

**Answer:** 48**12.**

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function  $f$  has zeros at  $x = -2$  and  $x = 5$ , and  $f(1) = -24$ . The function  $h$  is defined by  $h(x) = f(x+2) - 3f(-x)$ . What is the value of  $h(1)$ ?

1.  $f(x) = a(x+2)(x-5)$ ;  $f(1) = a(3)(-4) = -24 \Rightarrow a = 2$ , so  $f(x) = 2(x+2)(x-5)$ .
2. Substitute  $x = 1$  first:  $h(1) = f(3) - 3f(-1)$ .
3.  $f(3) = 2(5)(-2) = -20$  and  $f(-1) = 2(1)(-6) = -12$ .
4.  $h(1) = -20 - 3(-12) = -20 + 36 = 16$ .

**Answer:** 16

**13.**

SAT CHALLENGE · NO CALCULATOR NEEDED

In right triangle  $PQR$ ,  $\angle R = 90^\circ$  and  $\sin P + \sin Q = \frac{\sqrt{6}}{2}$ . What is the value of  $\tan P + \tan Q$ ?

1.  $\angle Q = 90^\circ - \angle P$ , so  $\sin Q = \cos P$  and  $\tan Q = \frac{1}{\tan P}$ .
2. Square the given sum:  $\sin^2 P + 2\sin P \cos P + \cos^2 P = \frac{6}{4} \Rightarrow 1 + 2\sin P \cos P = \frac{3}{2}$ .
3.  $\sin P \cos P = \frac{1}{4}$ .
4.  $\tan P + \tan Q = \frac{\sin P}{\cos P} + \frac{\cos P}{\sin P} = \frac{1}{\sin P \cos P} = 4$ .

**Answer:** 4**14.**

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function  $f$  has an axis of symmetry at  $x = 2$ , and its graph passes through  $(3, -3)$  and  $(5, 13)$ . The functions  $g$  and  $h$  are defined by  $g(x) = \frac{x-6}{2}$  and  $h(x) = 3 - f(g(x))$ . What is the  $y$ -coordinate of the  $y$ -intercept of the graph of  $y = h(x)$ ?

1. Vertex form with axis  $x = 2$ :  $f(x) = a(x-2)^2 + k$ .
2.  $f(3) = a + k = -3$  and  $f(5) = 9a + k = 13$ . Subtract:  $8a = 16 \Rightarrow a = 2, k = -5$ .
3.  $f(x) = 2(x-2)^2 - 5$ . The  $y$ -intercept of  $h$  is  $h(0)$ .
4. Inner value first:  $g(0) = -3$ . Then  $f(-3) = 2(-5)^2 - 5 = 45$ .
5.  $h(0) = 3 - 45 = -42$ .

**Answer:** -42

**15.**

SAT CHALLENGE · NO CALCULATOR NEEDED

The function  $f$  is defined by  $f(x) = c \cdot b^{x+3}$ , where  $c$  and  $b$  are positive constants. If  $f(1) = 1296$  and  $f(3) = 729$ , then for each increase of 1 in the value of  $x$ ,  $f(x)$  decreases by  $p\%$ . What is the value of  $p$ ?

1.  $f(1) = cb^4 = 1296$  and  $f(3) = cb^6 = 729$ .
2. Divide to cancel  $c$ :  $b^2 = \frac{729}{1296} = \frac{9}{16}$ . Since  $b > 0$ ,  $b = \frac{3}{4}$ .
3. Each unit increase in  $x$  multiplies  $f$  by  $\frac{3}{4}$ , a decrease of  $1 - \frac{3}{4} = \frac{1}{4} = 25\%$ .
4.  $p = 25$ . (The value of  $c$  is not needed.)

**Answer:** 25